

NLP Foundations

Module 6 Week A – Text Processing and Named Entity Recognition

Get to know Lana — Fun Fact

The Software Factory Intensive

A two-day hands-on workshop for software engineers building autonomous, agent-driven development pipelines. Pick your city.

📅 April 21st – 22nd

📍 Seattle, WA

👤 In-person

Event by

AI TINKERERS

| Hosted by

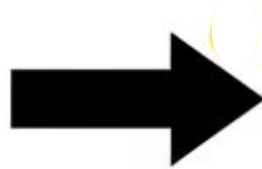
Actual AI

I'm learning to build "Software Factories"

- This week I completed a 2 day intensive training
- Gas City, Inc. framework
 - orchestration runtime - it runs agents, routes work, manages sessions
- 9 specialized agents working a single feature end to end using a pack designed by Actual AI
 - Planner → Architect → Designer → PM → Builder → Validator → Reviewer → Deployer
- Bringing the patterns to LevelUp's pipelines

From Tables To Text

customer_id	churn	monthly charges	monthly tenure
001	Yes	120	10
002	No	35	24
003	Yes	80	5
004	No	60	18



The United Nations announced new climate targets at COP28. Jordan committed to cutting emissions by 2030. Apple is investing in renewable energy.

One Sentence

**Apple is looking at buying
U.K. startup for \$1 billion.**

What Your Brain Did

Apple ORG is looking at buying U.K. GPE startup for \$1 billion MONEY .

ORG · organization
 GPE · geopolitical entity
 MONEY · amount
 action verb phrase
 target type

Named Entity Recognition

Apple is looking at
buying U.K. startup
for \$1 billion.



ORG

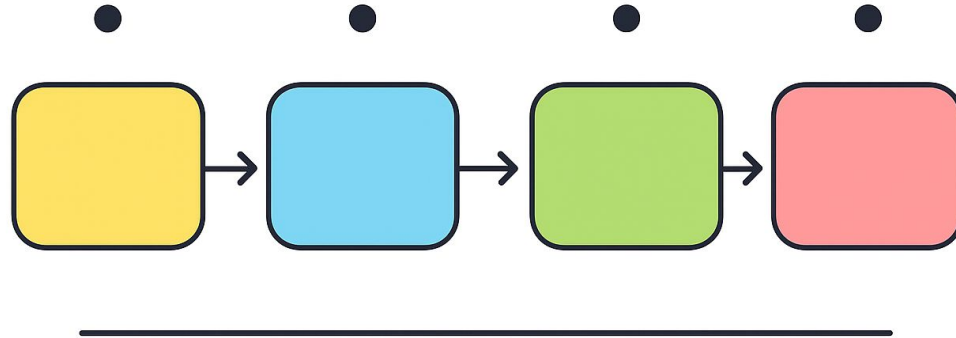


GPE



MONEY

Look at it like a Pipeline



raw text → structured output

Tokenization

Where do words begin and end.

Normalization

Different surface forms of the same word, should we treat them as the same.

POS tagging

What grammatical role does each word play.

NER

Which spans of text refer to real-world things, and what kind of thing.

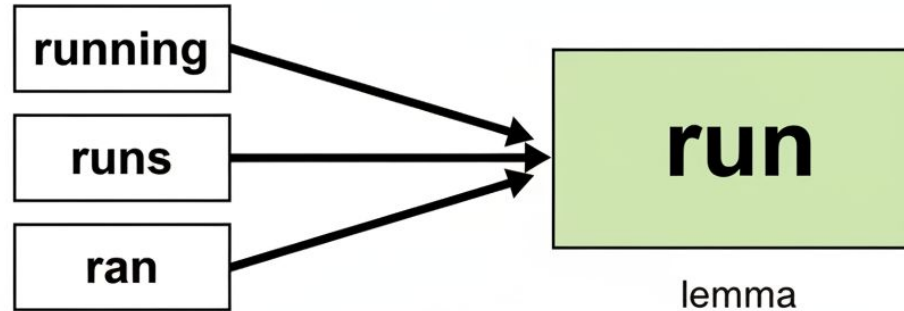
Tokenization

Apple is looking at U.K.



Apple is looking at U.K. .

Normalization



Stop Words

The cat sat on a mat in the sun.



STOP WORD REMOVAL: CROSS OUT THE STOP WORDS

~~The~~ **cat** **sat** ~~on a~~ **mat** ~~in the~~ **sun** .



WHAT REMAINS

cat sat mat sun

When Preprocessing Hurts

BEFORE

The United Nations **ORG** announced new climate targets.

↓ *apply standard preprocessing: lowercase + drop stop words ("the" goes)*

AFTER

united nations **?** announced new climate targets. *capitalization gone, leading "the" gone, entity weakened*

Capitalization Matters

US

GPE

"US raises tariffs on steel."



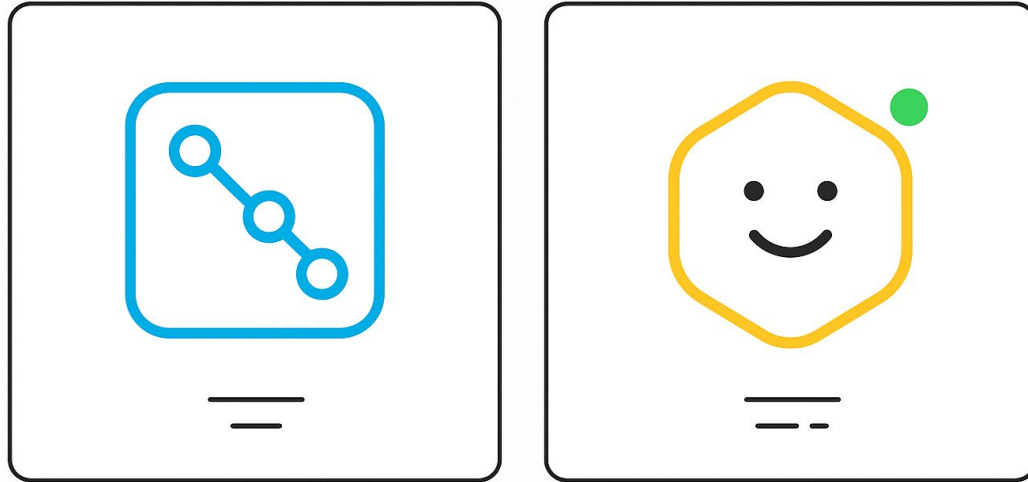
lowercase the text before NER — the entity signal is gone

us

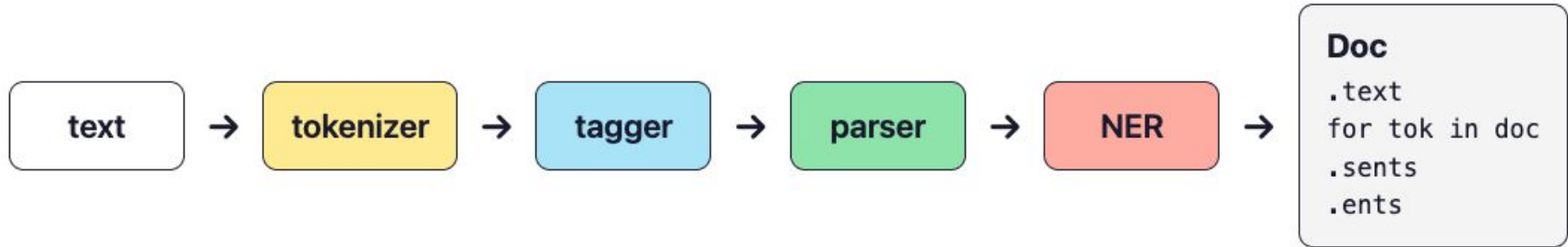
pronoun, not an entity

"Tell us about it."

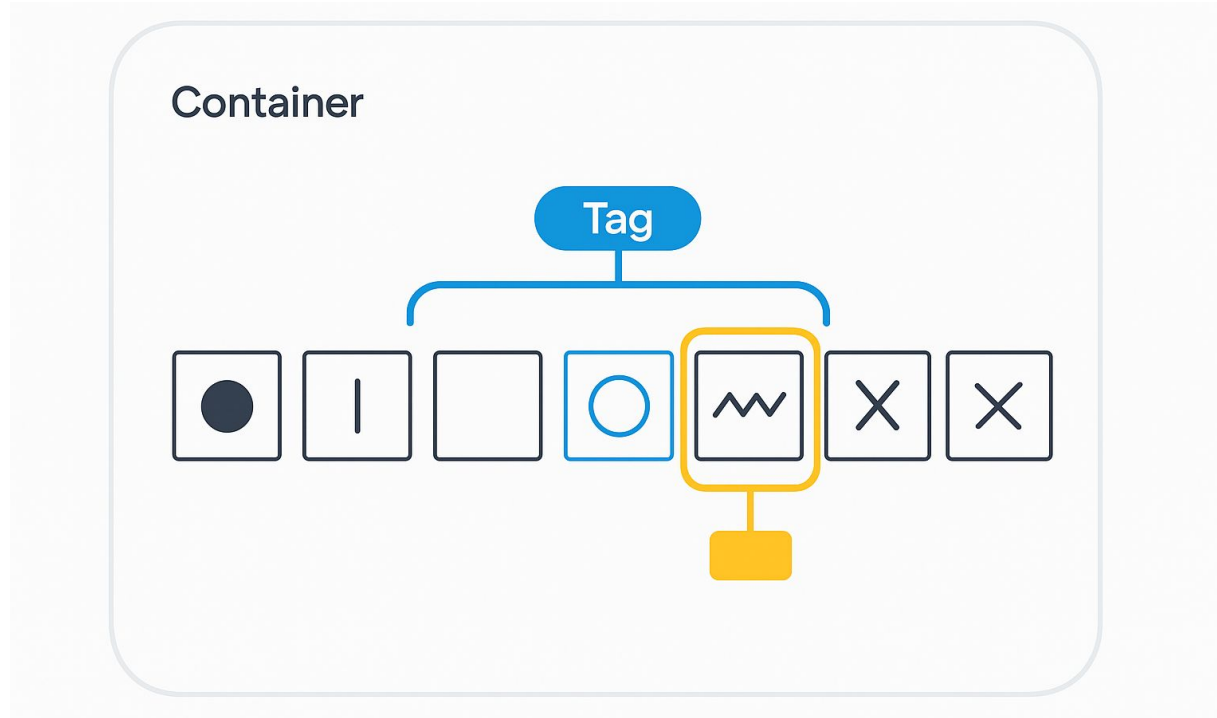
spaCy and Hugging Face



The spaCy Pipeline



Doc, Token, Span



Part-of-Speech Tags

Apple	is	looking	at	buying	U.K.	startup	for	\$	1	billion	.
PROPN	AUX	VERB	ADP	VERB	PROPN	NOUN	ADP	SYM	NUM	NUM	PUNCT

Entity Spans

WHAT DISPLACY RENDERS

Apple **ORG** is looking at buying U.K. **GPE** startup for \$1 billion **MONEY** .



WHAT YOUR CODE RETURNS (ONE ROW PER SPAN IN DOC.ENTS)

text	label_	start_char	end_char
"Apple"	ORG	0	5
"U.K."	GPE	27	31
"\$1 billion"	MONEY	44	54

displacy

Apple **ORG** is looking at buying U.K. **GPE** startup for \$1 billion **MONEY** . The United Nations **ORG** announced new climate targets at COP28 in Dubai **GPE** . Jordan **GPE** committed to cutting emissions by 30 percent **PERCENT** before 2030 **DATE** .

Hugging Face NER

SPACY

```
import spacy

nlp = spacy.load("en_core_web_sm")
doc = nlp("Apple is looking at buying U.K. startup for $1 billion.")

doc.ents
# (Apple, U.K., $1 billion)
```

returns: tuple of Span objects

HUGGING FACE

```
from transformers import pipeline

ner = pipeline("ner", model="dslim/bert-base-NER",
               aggregation_strategy="none") # token-level output
ner("Apple is looking at buying U.K. startup for $1 billion.")
# [{ 'entity': 'B-ORG', 'score': 0.998, 'word': 'Apple', 'start': 0, 'end': 5},
#   { 'entity': 'B-LOC', 'score': 0.991, 'word': 'U', 'start': 26, 'end': 27},
#   { 'entity': 'I-LOC', 'score': 0.987, 'word': '.', 'start': 27, 'end': 28},
#   { 'entity': 'I-LOC', 'score': 0.985, 'word': 'K', 'start': 28, 'end': 29},
#   { 'entity': 'I-LOC', 'score': 0.983, 'word': '.', 'start': 29, 'end': 30}]
# Note: $1 billion has no entry. CoNLL has no MONEY type, so the model
# leaves those tokens as 0 (outside any entity).
```

returns: list of dicts

Subword Tokens

RAW HF OUTPUT (3 TOKEN ROWS)



↓ merge subwords (drop ## prefix) + strip B-/I-

AFTER YOUR CODE CLEANS IT UP



spaCy vs Hugging Face

spaCy (en_core_web_sm)

trained on OntoNotes — about 18 entity types

Apple **ORG** is looking at buying U.K. **GPE** startup for \$1 billion **MONEY** .

Hugging Face (dslim/bert-base-NER)

trained on CoNLL — only PER, ORG, LOC, MISC

Apple **ORG** is looking at buying U.K. **LOC** startup for \$1 billion.

Same sentence. Two label schemas. U.K. is GPE in OntoNotes and LOC in CoNLL. CoNLL has no MONEY type, so \$1 billion is not labeled.

Unicode And Mojibake

UTF-8 FILE ON DISK

café

bytes: 0x63 0x61 0x66 0xC3 0xA9

↓ open the file telling Python it is Latin-1 (encoding mismatch)

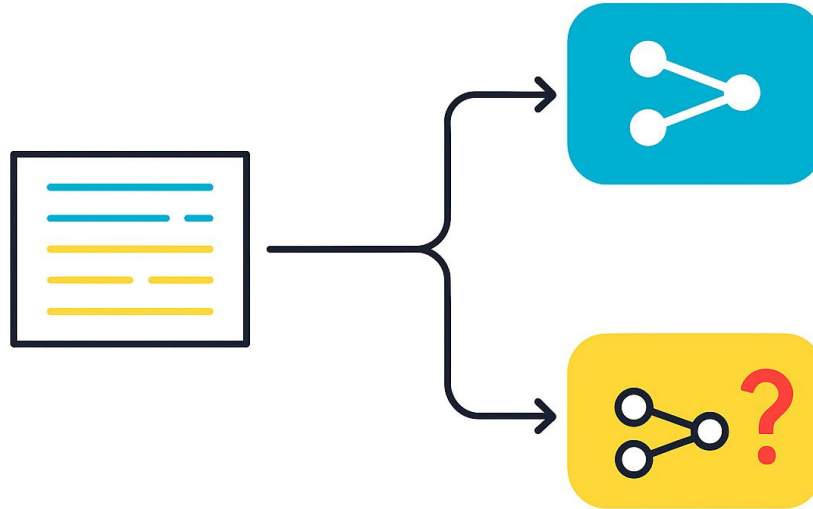
WHAT PYTHON DECODES

cafÃ©

same bytes, wrong rules: 0xC3 → Ã, 0xA9 → ©

Mojibake (Japanese: character transformation). The bytes did not change. The decoding rules did. Fix: specify `encoding="utf-8"` when you read the file.

Filter By Language First



Evaluating NER

	Predicted: Entity	Predicted: Not Entity
Gold: Entity	TP true positive	FN false negative
Gold: Not Entity	FP false positive	N/A no TN at entity level

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \cdot P \cdot R}{P + R}$$

A Worked Example

TP = 7 TP =	12 Predicted = 12	10 Gold = 10
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Precision = $7 / 12 \approx 0.58$	0.58
Recall = $7 / 10 = 0.70$	0.70
F1 ≈ 0.64	0.64

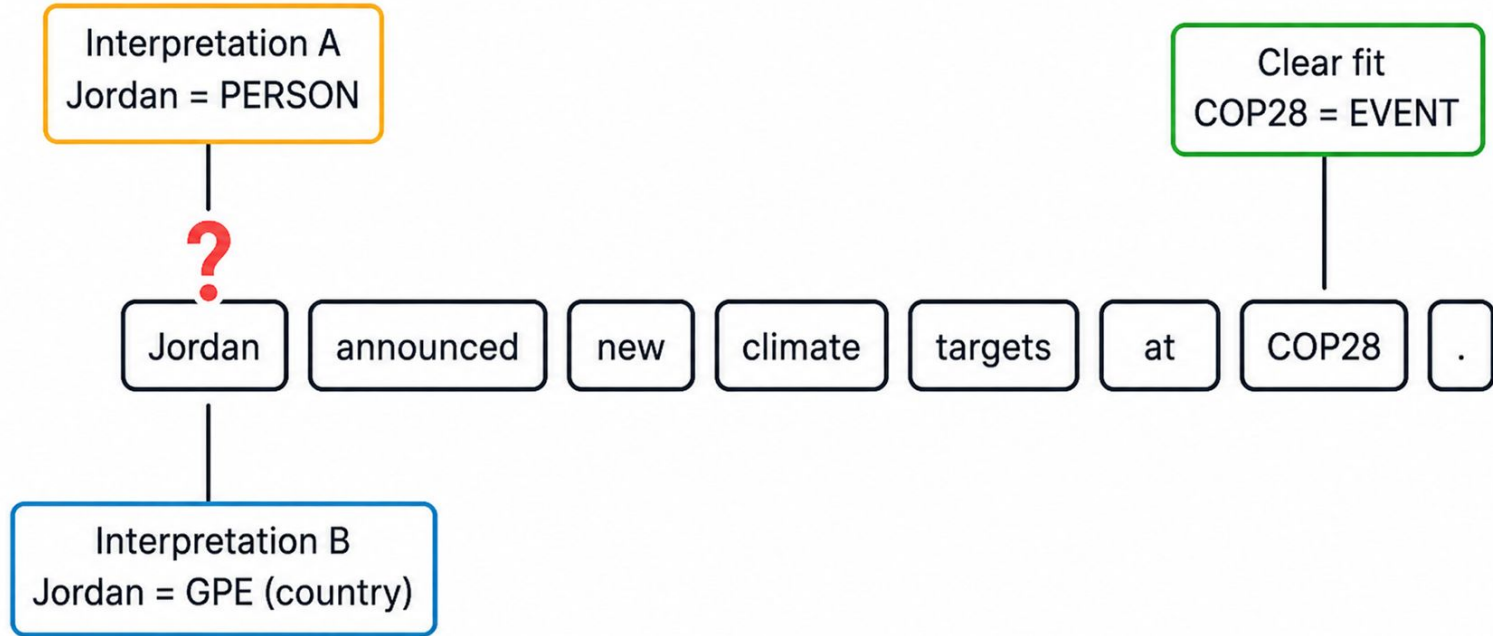
Pick A System

spaCy	
Precision	0.72
Recall	0.68
F1	0.70

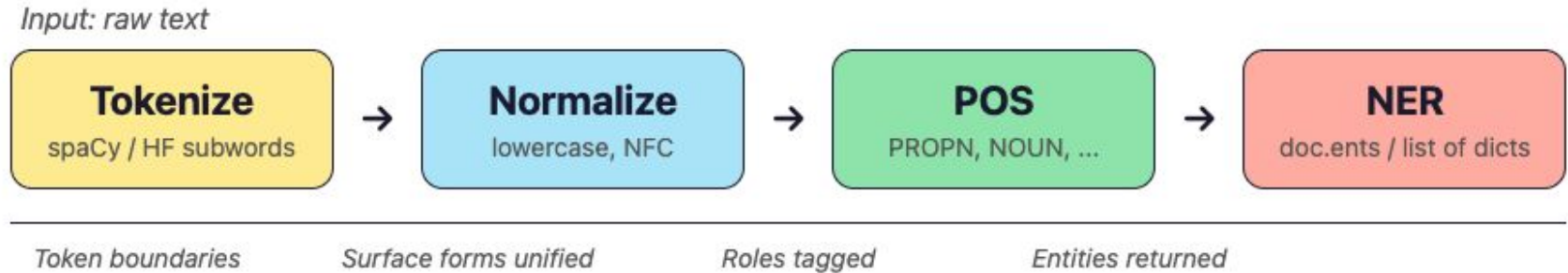
Hugging Face	
Precision	0.63
Recall	0.58
F1	0.60

Higher F1 on your gold subset → use **spaCy** on the integration

When NER Fails



NLP Pipeline Pattern



What's Next



Apple is looking at
buying **U.K.** startup
for **\$1 billion.**

ORG
GPE
MONEY

? Jordan announces
climate targets at COP28.

Questions

Announcements

Congratulations!



This week's Module Mentors

- Nayef
- Shahd
- Amer
- Bashar